SAFE RLHF VIA MINMAX PENALTIES: VALUE-FUNCTION-DERIVED SAFETY SIGNALS FOR LLM FINE-TUNING

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Abstract

Reinforcement Learning (RL) has demonstrated remarkable results, from mastering games like Atari and Go to aligning Large Language Models (LLMs) with human preferences via Reinforcement Learning from Human Feedback (RLHF). However, RLHF-trained models can still produce harmful content such as hate speech and discriminatory outputs, motivating a growing body of work on Safe RLHF. Existing safety alignment frameworks address this either through a KL divergence penalty, which constrains how far the policy drifts from a reference model at the cost of a manually tuned coefficient and does not provide a principled safety guarantee, or through a learned cost model, which requires additional harmlessness annotations and a Lagrange multiplier to balance the safety constraint against the helpfulness reward, introducing a second trained model and further hyperparameter burden. Recent work on the ROSARL algorithm shows that, in safe RL for robotic control, an agent's own value function can supply a self-calibrating safety penalty: a Minmax penalty derived from running estimates of the best and worst values the agent has encountered during training, without requiring an auxiliary cost model. This research investigates whether this Minmax penalty can be adapted to PPO-based LLM fine-tuning as the primary safety mechanism It asks whether doing so can reduce harmful outputs relative to a KL-penalty baseline without additional annotation or hyperparameter tuning. The proposed experiments evaluate this approach on open-source language models of varying scale, fine-tuned on a large-scale human-preference safety dataset and assessed against a KL-penalty baseline on held-out harm-rate benchmarks. More broadly, this research provides an accessible route to improving safety in RL-based LLM fine-tuning.

Declaration

I, Wiendy Maboa, hereby declare the contents of this proposal to be my own work. This proposal is submitted for the degree of Master of Science in the field of Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Maboa W.L

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August 7, 2026

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Contents

Chapter 1

Introduction

Reinforcement Learning (RL) has a history of exceeding its own expectations. For decades, its successes were confined to toy domains and simple simulations. That changed in 2015 when Mnih *et al.* [2015] showed that a single neural network, trained with RL from raw pixel inputs, could reach human-level performance across dozens of Atari games without any hand-engineered features. A year later, AlphaGo [Silver *et al.* 2016] defeated the European Go champion, solving a combinatorial problem thought to be decades from machine mastery. These milestones did more than set benchmarks: they established that deep RL could discover sophisticated strategies in high-dimensional spaces, and shifted the field's question from *whether* RL could match human performance to *which domains* it would transform next.

The answer, increasingly, is language. Christiano *et al.* [2017] showed that complex RL tasks could be solved from pairwise human preference labels without access to a true reward function. Ouyang *et al.* [2022] scaled this insight to GPT-3 [Brown *et al.* 2020], producing InstructGPT: a model that substantially outperformed its base counterpart on instruction-following by treating the Large Language Model (LLM) as an RL agent. In this paradigm, known as Reinforcement Learning from Human Feedback (RLHF), an incoming prompt is the state, token-by-token generation is a sequence of actions, and a reward model trained on human preferences provides the scalar feedback signal. RLHF has become the dominant paradigm for aligning LLMs with human intent [Lambert *et al.* 2023], and is typically optimised with Proximal Policy Optimisation (PPO) [Schulman *et al.* 2017].

Interestingly, this RL framing of language model alignment is not just a computational convenience. Neuroscience research on harm avoidance in biological agents offers a complementary perspective: Lockwood *et al.* [2020] showed that when learning to avoid harming others, humans prioritise model-free RL signals, the computationally efficient system that assigns values to actions through accumulated experience rather than deliberate reasoning. The *NeuroAI for AI Safety* roadmap [Mineault *et al.* 2024] formalises this intuition, arguing that studying how biological agents navigate environments while maintaining self-preservation is a productive route toward AI safety. We propose a Minmax penalty that operates on precisely this principle: a simple, reactive, value-based signal that does not require explicit reasoning about harm categories.

While RLHF is the standard paradigm for model alignment, it has introduced significant safety challenges, in part because safety training often fails to generalize as broadly as model capabilities [Shida *et al.* 2026; Dai *et al.* 2024; Casper *et al.* 2023]. Aligned models remain vulnerable to producing harmful content, including hate speech, discrimination, and instructions for self-harm, at fulfilment rates that vary markedly depending on the model's specific safety policy [Xie *et al.* 2025]. These persistent vulnerabilities, alongside the risk of reward hacking against imperfect proxy rewards even at inference time [Khalaf *et al.* 2025] and the risk of deceptive alignment, where a model appears compliant while pursuing misaligned goals internally [Ji *et al.* 2025], can severely degrade the trustworthiness and utility of LLMs in high-stakes deployment.. The established toolkit for constrained optimisation in RL [García and Fernández 2015; Achiam *et al.* 2017] has been adapted to address this, giving rise to the subfield of Safe RLHF, but the resulting approaches carry significant costs. This is worth distinguishing carefully from PPO's own clipped surrogate objective, which approximates a trust region purely for training stability and has no safety role. The KL divergence penalty referred to throughout the safe RLHF literature is a separate term, added to anchor the policy to a frozen reference model and discourage reward hacking [Ziegler *et al.* 2019; Ouyang *et al.* 2022]; it requires a manually tuned coefficient and introduces an implicit tradeoff between safety and capability. Learned cost models [Dai *et al.* 2024], building on constrained policy optimisation methods for safe RL [Achiam *et al.* 2017], train a separate model on human harmlessness labels and balance it against the helpfulness reward via a Lagrange multiplier: a second scalar that is itself learned during training to keep the cost model's constraint satisfied. HC-RLHF [Chittepu *et al.* 2025] adds high-confidence probabilistic guarantees but inherits the same infrastructure requirements. All of these approaches add components external to the core RL loop, introducing annotation cost, additional trained models, or extra hyperparameters, or some combination of the three. Table 1.1 summarises the structural differences.

Approach	**Extra model?**	**Extra annotation?**	**Extra safety hyperparameter?**	**Online?**
KL penalty baseline (standard RLHF) [Ziegler *et al.* 2019; Ouyang *et al.* 2022]	No	No	Yes (β)	Yes
Safe RLHF [Dai *et al.* 2024], adapting CPO [Achiam *et al.* 2017]	Yes	Yes	Yes (λ, d)	Yes
HC-RLHF [Chittepu *et al.* 2025]	Yes	Yes	Yes	Yes
RLHF + Minmax (ours)	**No**	**No**	**No**[1]	**Yes**

Table 1.1: Structural comparison of safe RLHF approaches. The Minmax penalty is the only method requiring no extra model, no extra annotation, and no new safety-specific hyperparameters.

This raises the central question of this proposal: can we replace KL divergence penalties and learned cost models with a theoretically motivated safety signal derived directly from the agent's own value function, without introducing new hyperparameters or auxiliary models?

[1]The Minmax penalty itself introduces no new hyperparameters: its bounds are derived from the agent's own training trajectory rather than set by hand. The small KL anchor retained in our implementation is a conservatism term inherited from the PPO training loop to prevent mode collapse, not a safety-specific parameter, and the unsafe-detection threshold is a property of the detector rather than of the Minmax mechanism itself; both are described in full in Chapter 4.

The ROSARL algorithm [Nangue Tasse *et al.* 2024] answers this in the affirmative for robotic control tasks. The agent tracks running estimates of the smallest and largest rewards it has observed, alongside corresponding value bounds derived jointly from those reward estimates and the critic's own value predictions. When an unsafe-state detector fires, the reward at that step is replaced by the gap between the lowest and highest of these value bounds, a quantity that is guaranteed to be non-positive and represents the worst possible value any reachable state could take, given everything the agent has observed so far. As training progresses and the agent explores more of the reward landscape, this penalty becomes monotonically more severe, providing an increasingly strong corrective signal without any manual tuning. The full update rule is given as Algorithm 1 in Chapter 4.

This proposal also connects to the emerging paradigm of RL with verifiable rewards (RLVR) [Lambert *et al.* 2024; Mroueh 2025; Su *et al.* 2025], where reward signals are computed deterministically rather than approximated by a learned model. The Minmax penalty fits this paradigm: it is derived directly from the critic's own value estimates, not from an opaque learned cost function. Where recent work asks whether verifiable rewards preserve safety alignment [Cho *et al.* 2025], this proposal investigates the complementary direction: using a value-function-derived signal to actively *improve* it.

Because Algorithm 1 only acts when the unsafe-state detector fires, detector quality is expected to be the primary variable affecting coverage. This motivates a two-phase research programme. In Phase 1, Detoxify [Hanu and Unitary team 2020] serves as the unsafe-state detector, providing a fast, reproducible safety signal for harm categories that manifest as surface-level toxic language. In Phase 2, Detoxify is replaced by an activation-based probe [Rateike *et al.* 2023; Han *et al.* 2025], enabling Algorithm 1 to intercept safety-relevant states before the output token is produced, and covering categories that output-level classifiers miss, such as self-harm. This two-phase architecture is illustrated in Chapter 4.

The contributions of this study are as follows.

1. **Contribution 1.** First application of the ROSARL Minmax penalty to PPO-based LLM fine-tuning. The study will evaluate this under multiple base models (GPT-2 small and a Qwen model at 1 to 3B parameters) and fine-tuning strategies (full fine-tuning and LoRA [Hu *et al.* 2022]), isolating the benefit to the Minmax mechanism rather than any specific model or training configuration.

2. **Contribution 2.** A proposed bridge between Algorithm 1 and activation-based safety detection [Han *et al.* 2025; Rateike *et al.* 2023]. Following recent follow-up work to Rateike *et al.* [2023] (currently under review) identifying hallucination detection as the most tractable initial subset for activation-level probes, the probe will first target factually incorrect and internally inconsistent outputs before extending to broader harm categories.

3. **Contribution 3.** A theoretical and empirical analysis of the Minmax penalty as a verifiable safety signal, situating it within the RLVR literature [Lambert *et al.* 2024; Mroueh 2025] and the safety-capability tradeoff debate [Cho *et al.* 2025].

Chapter 2 provides background and related work. Chapter 3 states the research questions and hypotheses. Chapter 4 describes the methodology. Chapter 5 outlines the research plan. Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter reviews the technical foundations required to situate the proposed research. Section 2.1 covers RL fundamentals and the RLHF pipeline. Section 2.2 reviews the safe RL literature. Section 2.3 introduces the RLVR paradigm. Section 2.4 describes the ROSARL algorithm in detail. Section 2.5 covers the datasets used. Section 2.6 reviews activation-based safety detection, which forms the basis of Phase 2.

2.1 Reinforcement Learning and RLHF

RL models sequential decision-making as a Markov Decision Process (MDP), defined as a tuple (S, A, P, R, γ), where S is the state space, A the action space, P the transition dynamics, R the reward function, and γ the discount factor. The agent learns a policy π maximising expected cumulative discounted reward [Puterman 2014; Sutton and Barto 2018]:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right] \tag{2.1}$$

PPO [Schulman *et al.* 2017] is the standard RL algorithm for RLHF. It stabilises training via a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big] \tag{2.2}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio and $\hat{A}_t$ is the advantage estimate. PPO maintains separate actor and critic networks; the critic provides value estimates $V(s)$ that are central to Algorithm 1.

The modern RLHF pipeline trains a reward model on human preference labels and uses PPO to fine-tune the LLM to maximise this reward subject to a KL divergence penalty:

$$R_{\text{RLHF}}(s, a) = r_\phi(s, a) - \beta\,\text{KL}[\pi_\theta(\cdot|s)\,\|\,\pi_{\text{ref}}(\cdot|s)] \tag{2.3}$$

where r_ϕ is the reward model, π_{ref} is the frozen reference policy, and β is the KL coefficient. This KL term, first introduced by Ziegler *et al.* [2019] and scaled to production by Ouyang *et al.* [2022] with InstructGPT, is the primary safety mechanism in standard RLHF: a regulariser rather than a principled safety guarantee. The limitations and historical risks of this framing are discussed by Lambert *et al.* [2023], including the tendency of reward models to encode annotation artefacts, and the lack of a principled basis for setting β.

Safe RLHF [Dai *et al.* 2024] extends this with a separately trained cost model and Lagrangian constrained optimisation. A complementary line of work, Constitutional AI [Bai *et al.* 2022], scales safety alignment by replacing some human feedback with AI-generated critiques and revisions guided by a written set of principles, reducing annotation cost at the price of a second model in the training loop:

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_C(\pi) \le d \tag{2.4}$$

where $J_C(\pi)$ is the expected cost from the cost model and d is a safety threshold. HC-RLHF [Chittepu *et al.* 2025] adds high-confidence constraint satisfaction, representing the current state of the art. Both require multi-GPU infrastructure, a second trained model, and annotated harmlessness labels.

2.2 Safe Reinforcement Learning

Safe RL [García and Fernández 2015] addresses the challenge of learning policies that maximise reward while satisfying behavioural constraints during training and deployment. Figure 2.1 illustrates the main families of approaches.

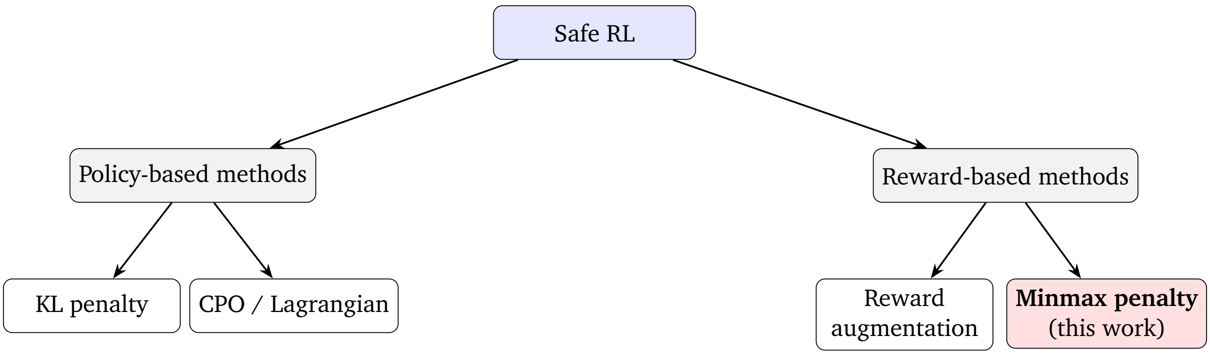

Figure 2.1: Taxonomy of safe RL approaches [García and Fernández 2015]. The Minmax penalty is a reward-based method requiring no learned cost model.

Constrained Policy Optimisation [Achiam *et al.* 2017] provides the first policy gradient method with formal worst-case constraint guarantees:

$$\pi_{k+1} = \arg\max_{\pi} J_R(\pi) \quad \text{s.t.} \ \ J_{C_i}(\pi) \le d_i \ \forall i, \quad D_{\text{KL}}(\pi \| \pi_k) \le \delta \tag{2.5}$$

Safe RLHF adapts CPO to the RLHF setting with a learned cost model. Distributional approaches [Yang *et al.* 2022] further strengthen safety by controlling tail-risk via CVaR. These methods provide the theoretical comparators that motivate asking whether any constraint model is necessary at all.

2.3 Reinforcement Learning with Verifiable Rewards

A parallel strand of recent work reframes reward design around verifiability: using signals that can be checked exactly rather than approximated by a learned model [Lambert *et al.* 2024; Mroueh 2025; Su *et al.* 2025]. A prominent example is Group Relative Policy Optimisation (GRPO), introduced by Shao *et al.* [2024] in DeepSeekMath, which replaces the learned value baseline with group-relative advantages computed from verifiable mathematical rewards. This eliminates reward model error and reduces reward hacking. Cho *et al.* [2025] study whether such training breaks safety alignment, finding that KL constraints can preserve safety guardrails. This proposal investigates the complementary direction: using a deterministically computed, verifiable value-function signal to actively improve safety rather than merely preserve it.

2.4 The ROSARL Algorithm

ROSARL [Nangue Tasse *et al.* 2024] proposes that the agent's own value function contains sufficient information to construct a principled safety penalty without auxiliary models. At each training step the algorithm maintains R_{min} and R_{max} as the running minimum and maximum of all rewards observed so far, and updates value bounds V_{min} and V_{max} via a three-way minimum and maximum that incorporates both the running reward bounds and the critic's current value estimate $V(s_t)$: $V_{\text{min}} \leftarrow \min(V_{\text{min}}, R_{\text{min}}, V(s_t))$ and $V_{\text{max}} \leftarrow \max(V_{\text{max}}, R_{\text{max}}, V(s_t))$. This anchoring ensures V_{min} tracks the worst value the agent could plausibly experience and V_{max} tracks the best, so that the penalty gap grows as the agent discovers the full range of achievable values. When the unsafe-state detector z_t fires, the reward is replaced with $R_{\text{unsafe}} = V_{\text{min}} - V_{\text{max}}$, which is always non-positive and lower than the value of any reachable state under the observed reward and value range. As the agent encounters wider extremes, R_{unsafe}

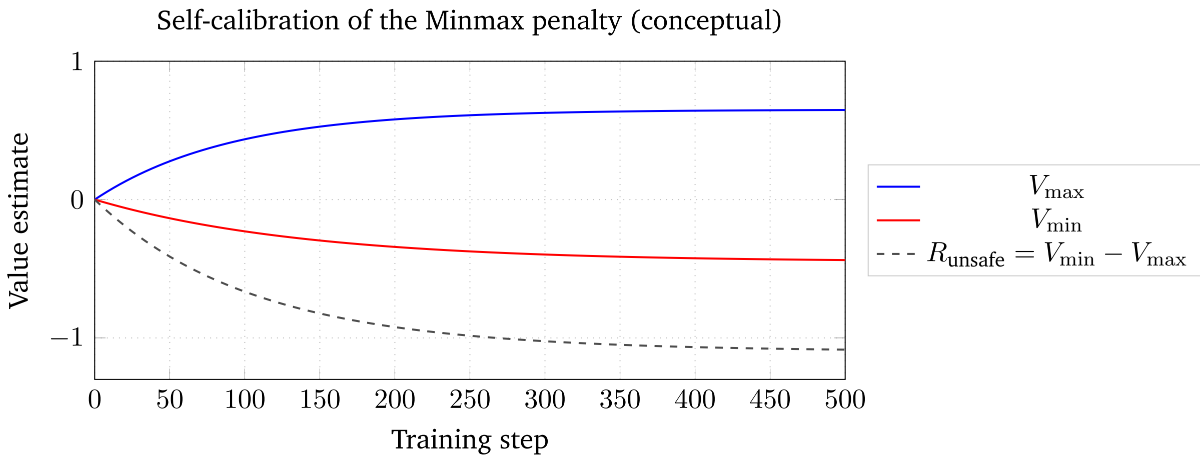

Figure 2.2: Conceptual illustration of the self-calibration property. As training progresses R_{unsafe} becomes monotonically more negative, providing increasingly strong corrective signal without any manual tuning.

becomes monotonically more negative, making the penalty more severe without any manual tuning. Figure 2.2 illustrates this self-calibration property.

ROSARL was validated on Safety-Gymnasium benchmarks (PointGoal, PointPush, HumanoidVelocity, PILLAR) and matches or outperforms Lagrangian, CPO, and Sauté-RL [Sootla *et al.* 2022] baselines under environmental stochasticity. This proposal is the first application to LLM fine-tuning.

2.5 Datasets

This section describes the datasets used for training and evaluation in this research.

BeaverTails [Ji *et al.* 2023] is a large-scale human preference dataset covering 14 harm categories: animal abuse, child abuse, controversial topics, discrimination, drug abuse, financial crime, hate speech, human rights violations, non-violent unethical behaviour, privacy violations, sexually explicit content, terrorism, violence, and weapons. Uniquely, it provides separate annotations for helpfulness and harmlessness for the same prompt. The training split (27,186 prompts) is used for fine-tuning; the held-out BeaverTails-Evaluation split (700 prompts) serves as the primary evaluation benchmark, shared with Safe RLHF and HC-RLHF to enable direct SOTA comparison.

PKU-SafeRLHF [Ji *et al.* 2024] extends BeaverTails with multi-level safety labels across 19 harm categories and three severity levels, providing paired safe and unsafe responses for the same prompts. This structure is directly relevant to the Phase 2 extension: the paired responses provide explicit positive and negative training examples for the activation probe.

HarmBench [Mazeika *et al.* 2024] provides over 500 unique adversarial harmful behaviours designed to probe LLM safety under worst-case conditions. Unlike BeaverTails, which covers naturalistic prompts, HarmBench is specifically designed for red-teaming: it includes standard, contextual, and multi-turn behaviours, as well as attacks that span multiple harm categories simultaneously. In this research, HarmBench serves as an out-of-distribution robustness evaluation, testing whether the Minmax penalty generalises beyond the harm categories and prompt styles seen during BeaverTails training. Full HarmBench evaluation is planned as a later milestone pending larger model access.

2.6 Activation-Based Safety Detection

This section reviews the activation-based safety detection literature that motivates Phase 2 of this research.

Rateike *et al.* [2023] demonstrate that LLMs encode unsafe information in internal activation vectors during generation, before the output token is produced. Using DeepScan (subset scanning), they identify specific neuron subsets in the final layers of OPT-6.7B that are anomalously active when the model is about to hallucinate. Crucially, hallucination detection was found to be the most tractable starting subset for detection, a finding from recent follow-up work currently under review that directly informs the

Phase 2 design: the activation probe will target factually incorrect outputs first before extending to broader harm categories. Cintas *et al.* [2025] extend this to persona representations in decoder-only LLMs, finding that safety-relevant representations are localised in the final third of layers, with the open question of whether these activations can steer generation.

SafeSwitch [Han *et al.* 2025] provides the most direct empirical precedent for Phase 2: internal activation signals can directly steer unsafe behaviour in deployed LLMs, achieving large reductions in harmful outputs while preserving utility. SafetyNet [Chaudhary and Barez 2025] provides complementary evidence by monitoring deceptive behavioural signals in activations in real time. Nguyen *et al.* [2025] further show that distributional surgery on activations can both detect and mitigate undesirable content layerwise.

Chapter 3

Research Questions

This chapter states the problem addressed by the research and the questions and hypotheses that structure the experimental programme.

3.1 Problem Statement

LLMs fine-tuned with PPO can produce harmful outputs at deployment at rates that undermine their trustworthiness in safety-critical applications [Dai *et al.* 2024; Chittepu *et al.* 2025]. Current mitigation strategies each introduce a structural cost: KL divergence penalties require manual tuning of the KL coefficient and trade safety against capability without a principled basis for setting the trade-off; learned cost models require a second trained model, additional harmlessness annotations, and a Lagrangian multiplier whose update dynamics introduce further hyperparameter burden. These mechanisms add components external to the core PPO loop, increasing engineering and annotation cost without providing a principled safety guarantee. This research addresses the question of whether the value function already present in the PPO training loop can itself supply a self-calibrating safety signal that replaces both mechanisms. The methodology in Section 4.1 operationalises this question through Algorithm 1, ablation studies that isolate the Minmax mechanism, and a two-phase pipeline that varies only the unsafe-state detector.

3.2 Research Hypothesis

This research investigates whether the ROSARL Minmax penalty can serve as the primary safety mechanism in PPO-based LLM fine-tuning, replacing KL divergence penalties and learned cost models, across multiple base models and fine-tuning strategies. The experimental programme is structured around three hypotheses; each hypothesis maps directly to one dimension of the ablation design described in Section 4.1.4, so that the three together cover the safety mechanism, the detector, and the architectural extension to activation-level detection.

- **H1 (mechanism).** A policy trained with the Minmax penalty produces fewer harmful outputs than a policy trained with KL divergence alone, measured by harm rate on BeaverTails-Evaluation. This effect is expected to persist across base models (GPT-2 small and a small Qwen model) and fine-tuning strategies (full fine-tuning and LoRA), isolating the mechanism as the source of the benefit rather than any model-specific property.
- **H2 (detector quality bounds coverage).** The Minmax penalty's harm reduction is bounded by the recall of the unsafe-state detector. Categories where the detector signal is weak or absent, such as self-harm under Detoxify, are not expected to improve and may regress, because Algorithm 1 only acts when $z_t = 1$. This frames detector quality as the primary variable modulating coverage, not the penalty mechanism itself.
- **H3 (consequence of H2).** If H2 holds, then replacing Detoxify with an activation-level probe should extend coverage to categories where Detoxify's recall is near zero (most notably self-harm), without altering the underlying Minmax mechanism. H3 tests this prediction directly: the penalty is held fixed, only the source of z_t is varied, so that any change in coverage is attributable to the detector rather than to a different safety algorithm.

Chapter 4

Methodology

This chapter describes the full experimental design, implementation, evaluation strategy, ablation studies, and limitations.

4.1 Experimental Methodology

This chapter details the experimental methodology. PPO is used for all fine-tuning runs, comparing a KL-penalty baseline against the Minmax penalty. The unsafe-state detector is Detoxify in Phase 1 and an activation-based probe in Phase 2. Evaluation is performed on two benchmarks: BeaverTails-Evaluation [Ji *et al.* 2023] as the in-distribution harm-rate measure, and HarmBench [Mazeika *et al.* 2024] as an out-of-distribution adversarial robustness measure. Ablations isolate the effect of the Minmax mechanism from confounds such as base model, fine-tuning strategy, and detector threshold. The methodology is structured around four components: data collection (Section 4.1.1), the two-phase pipeline architecture (Section 4.1.2), ablation studies (Section 4.1.4), and limitations and constraints (Section 4.1.5).

4.1.1 Data Collection

To compare the Minmax penalty with existing methods, PPO fine-tuning with the Minmax penalty is implemented on GPT-2 small and a small Qwen model. The training split of BeaverTails is used for fine-tuning, with evaluation on the held-out BeaverTails-Evaluation split and on HarmBench.

Implementation framework. The primary implementation framework is the TRL library's `PPOTrainer` [von Werra *et al.* 2020], into which the Minmax penalty is integrated as a reward wrapper following the structure of Algorithm 1. The PKU-SafeRLHF codebase [Ji *et al.* 2024] is used as the reference implementation for understanding the Safe RLHF baseline design, not as the running framework: it requires DeepSpeed, which does not build on the available Windows hardware. A later milestone will port the Minmax wrapper into the PKU-SafeRLHF codebase on a Linux cluster to enable direct head-to-head comparison with the published Safe RLHF and HC-RLHF baselines under matched infrastructure.

Algorithm 1 Implementation. The Minmax penalty is implemented as a reward wrapper around the TRL `PPOTrainer` [von Werra *et al.* 2020]. The full procedure is given in Algorithm 1.

Algorithm 1 PPO + Minmax Penalty (LLM adaptation of ROSARL [Nangue Tasse *et al.* 2024])

Require: LLM π_θ, critic V_ψ, unsafe detector $\mathcal{D}$, threshold τ, KL anchor β

1: Initialise $R_{\min} \leftarrow 0,\ R_{\max} \leftarrow 0,\ V_{\min} \leftarrow 0,\ V_{\max} \leftarrow 0$ ▷ as in original ROSARL
2: **for** each training step t **do**
3:     Sample prompt $x \sim \mathcal{X}_{\text{train}}$
4:     Generate response $y \sim \pi_\theta(\cdot \mid x)$
5:     $\delta_t \leftarrow \mathcal{D}(y)$ ▷ detector score
6:     $r_t^{\text{raw}} \leftarrow -\delta_t$ ▷ raw reward (LLM adaptation: negated Detoxify score)
7:     $V_t \leftarrow V_\psi(x, y)$ ▷ critic value estimate
8:     $R_{\min} \leftarrow \min(R_{\min},\ r_t^{\text{raw}});\quad R_{\max} \leftarrow \max(R_{\max},\ r_t^{\text{raw}})$ ▷ ROSARL reward bounds
9:     $V_{\min} \leftarrow \min(V_{\min},\ R_{\min},\ V_t);\quad V_{\max} \leftarrow \max(V_{\max},\ R_{\max},\ V_t)$ ▷ ROSARL three-way value bounds
10:     **if** $\delta_t > \tau$ **then** ▷ unsafe state detected
11:         $r_t \leftarrow \max(V_{\min} - V_{\max},\ -1.0)$ ▷ Minmax penalty; -1.0 floor is a stabilisation modification, not in original ROSARL
12:     **else**
13:         $r_t \leftarrow r_t^{\text{raw}}$ ▷ LLM adaptation: detector reward (original uses environment reward)
14:     **end if**
15:     Update π_θ, V_ψ via PPO with r_t and KL penalty β
16: **end for**

Key design choices, with explicit attribution to either ROSARL or this work:

- **Reward-bound tracking ($R_{\min}$, $R_{\max}$) and the $R_{\text{unsafe}} = V_{\min} - V_{\max}$ formula** are taken unchanged from ROSARL [Nangue Tasse *et al.* 2024].
- **Three-way value-bound update.** Following Algorithm 1 of Nangue Tasse *et al.* [2024], $V_{\min}$ and $V_{\max}$ are updated using a three-way minimum and maximum respectively: $V_{\min}$ incorporates both the running minimum reward $R_{\min}$ and the critic's current value estimate $V(s_t)$, ensuring the penalty gap grows as the agent discovers the full range of achievable values. $V_{\max}$ is correspondingly anchored to the highest reward and value seen, providing an optimistic upper bound that makes R_{unsafe} conservative by construction.
- **The -1.0 floor on the unsafe reward** is not part of the original Algorithm 1; it was introduced as an engineering modification to guard against catastrophic PPO ratio violations during early training, when value estimates are not yet well-calibrated and $V_{\min} - V_{\max}$ can briefly take very large negative values.

- **The safe-branch reward** $r_t \leftarrow -\delta_t$ is this work's LLM-setting adaptation: in the absence of a dense environment reward signal, the negated Detoxify toxicity score provides a continuous signal that the policy can optimise, encouraging progressively less toxic outputs even when no unsafe flag is triggered. This differs from the original Algorithm 1, in which the environment reward is passed through unchanged for safe transitions.
- **The small KL anchor (**$\beta = 0.01$**)** is retained solely to prevent mode collapse and is distinct from the $\beta = 0.2$ used in the KL baseline condition.
- **Reducing PPO epochs from 4 to 1** prevents the probability ratio from exploding when the reward sign flips abruptly between the safe and unsafe branches.

Base models and fine-tuning strategies. GPT-2 small (117M parameters) serves as the primary base model, feasible on the available hardware. A Qwen model in the 1 to 3B parameter range is planned for the model-scale ablation (Ablation 2), to be run on a compute cluster. LoRA [Hu *et al.* 2022] is used as the fine-tuning strategy for larger models, as it reduces memory requirements while still permitting comparison across fine-tuning methods in Ablation 3. In Phase 1, Detoxify [Hanu and Unitary team 2020] provides the binary signal z_t indicating whether the model output is classified as harmful. In Phase 2, Detoxify is replaced with an activation-based probe targeting hallucination detection as the initial subset, following the methodology of Rateike *et al.* [2023].

Evaluation benchmarks. The primary evaluation benchmark is the held-out BeaverTails-Evaluation set (700 prompts across 14 harm categories). Both trained models generate 32 tokens per prompt; outputs are classified as harmful or benign, and harm rate is reported in aggregate and by category, enabling direct comparison with published Safe RLHF [Dai *et al.* 2024] and HC-RLHF [Chittepu *et al.* 2025] results, which use the same split. The secondary benchmark is HarmBench [Mazeika *et al.* 2024], which provides over 500 adversarial behaviours specifically designed for red-teaming, spanning standard, contextual, and multi-turn attacks that cross harm categories. Evaluation on HarmBench tests whether the Minmax penalty generalises beyond the naturalistic BeaverTails prompts seen during training. Following Mazeika *et al.* [2024], attack success rate (ASR) is the primary metric, reported overall and broken down by attack family. HarmBench evaluation is planned as a later phase, after Phase 1 results are stable.

4.1.2 Pipeline Implementation: Two-Phase Architecture

The research is structured in two phases, each defined by the unsafe-state detector that feeds the binary signal z_t into Algorithm 1. In both phases the PPO training loop and the Minmax penalty itself are identical; only the source of z_t changes. The two phases are illustrated in Figure 4.1 and Figure 4.2.

Phase 1: Detoxify as the unsafe-state detector. At each training step a prompt x is drawn from BeaverTails and the policy π_θ generates a response y. Detoxify [Hanu and

Unitary team 2020] scores y and returns a toxicity probability $\delta_t \in [0, 1]$; the binary signal is z_t. The Minmax wrapper then sets the per-trajectory reward to $R_{\text{unsafe}} = V_{\text{min}} - V_{\text{max}}$ whenever $z_t = 1$, and to the standard reward $-\delta_t$ otherwise, before passing it to the PPO update. This phase is the focus of the empirical evaluation in this proposal: it requires no additional training of a safety model, and produces a fast feedback loop that makes the core Minmax claim falsifiable on BeaverTails-Evaluation and HarmBench. Figure 4.1 shows the Phase 1 architecture.

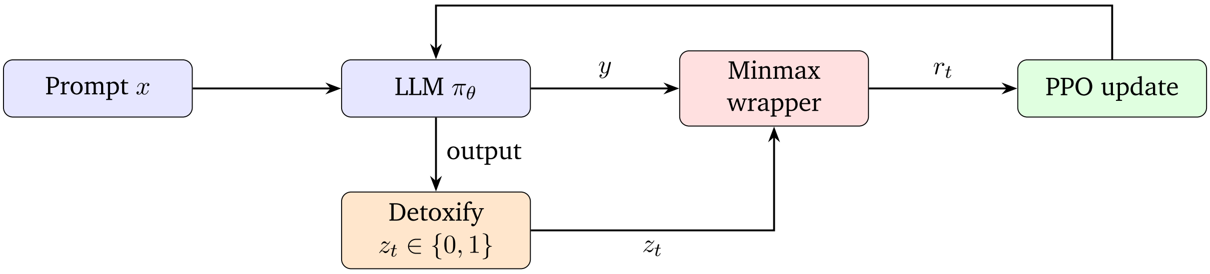

Figure 4.1: Phase 1 architecture. Detoxify classifies the model output, providing the binary unsafe flag z_t to the Minmax wrapper. When $z_t = 1$, the wrapper replaces the reward with $R_{\text{unsafe}} = V_{\text{min}} - V_{\text{max}}$ before passing it to the PPO update.

Phase 2: Activation probe as the unsafe-state detector. Phase 2 keeps the rest of the pipeline fixed and replaces Detoxify with a lightweight activation probe trained on internal model states, following Rateike *et al.* [2023]; Han *et al.* [2025]. The motivation is that surface-level classifiers cannot detect categories such as self-harm, where the output is fluent and non-toxic but the underlying intent is unsafe. Phase 2 is a planned extension scoped as a single milestone; a detailed design is deferred until Phase 1 results are stable. Figure 4.2 shows the Phase 2 architecture.

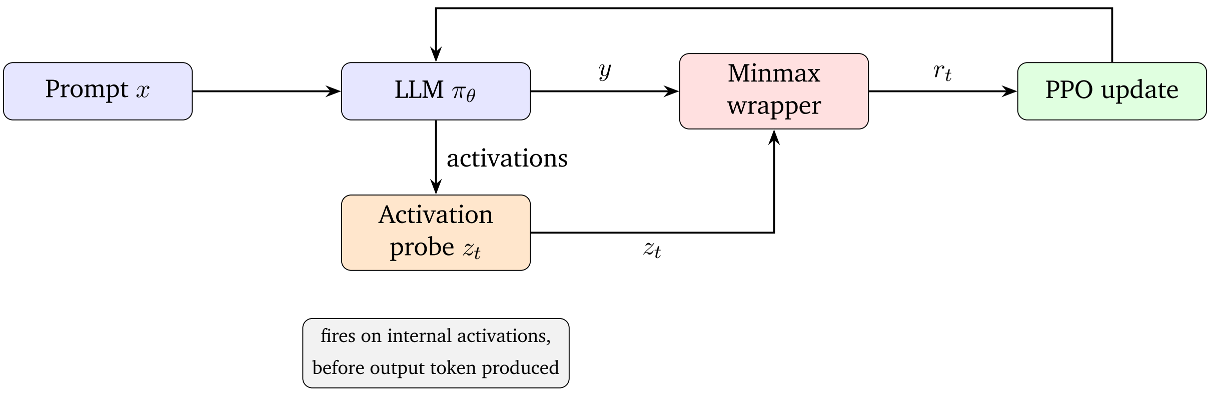

Figure 4.2: Phase 2 architecture. The activation probe replaces Detoxify, detecting unsafe states from internal activations before the output token is produced. This extends Algorithm 1's coverage to harm categories that surface-level classifiers miss.

Evaluation across both phases. Trained policies are evaluated on two complementary benchmarks. The primary, in-distribution benchmark is BeaverTails-Evaluation: 700

held-out prompts across 14 harm categories. Both trained models generate 32 tokens per prompt; outputs are classified as harmful or benign, and harm rate is reported in aggregate and per category. Using the same split as Safe RLHF [Dai *et al.* 2024] and HC-RLHF [Chittepu *et al.* 2025] enables direct comparison with published results.

The secondary, out-of-distribution benchmark is HarmBench [Mazeika *et al.* 2024], which provides over 500 adversarial behaviours designed for red-teaming, spanning standard, contextual, and multi-turn attacks that cross harm categories. HarmBench tests whether the Minmax penalty generalises beyond the naturalistic BeaverTails prompts seen during training, and is the stronger test of H_1: a method that only reduces harm on in-distribution prompts could simply be reproducing the training distribution rather than learning a robust safety policy. Following Mazeika *et al.* [2024], attack success rate (ASR) is the primary metric, reported overall and broken down by attack family. Full HarmBench evaluation is run after Phase 1 results are stable, and again after Phase 2.

At each training step the following are logged: mean reward, V_{min}, V_{max}, R_{unsafe}, policy KL divergence, and PPO ratio statistics, enabling inspection of the self-calibration dynamics predicted by Nangue Tasse *et al.* [2024]. Ablations on safety mechanism, base model, fine-tuning strategy, and detector threshold isolate the contribution of the Minmax mechanism from confounds (see Section 4.1.4).

4.1.3 Evaluation Metrics and Baselines

This section specifies the metrics used to evaluate each experimental condition and the baseline(s) each ablation is compared against. Metric choice follows prior safe RLHF work where possible, to keep results comparable to published numbers.

Evaluation Metrics

Harm rate. The primary metric throughout this thesis, following Safe RLHF [Dai *et al.* 2024] and HC-RLHF [Chittepu *et al.* 2025]: the proportion of generated responses on BeaverTails-Evaluation classified as harmful by an independent judge (Detoxify in Phase 1, an activation probe in Phase 2). Harm rate is reported both in aggregate and per harm category, since per-category breakdowns are what expose detector coverage gaps.

Harm rate, non-empty responses. Empty or near-empty generations are scored separately and excluded from the harmful/benign denominator, since an empty response trivially registers as non-toxic and would otherwise deflate the reported harm rate without reflecting a genuine safety improvement. Both $\text{harm\_rate}_{\text{all}}$ and $\text{harm\_rate}_{\text{nonempty}}$ are reported, with the latter treated as primary.

Response diversity. The proportion of unique responses across the evaluation set (unique strings / total prompts) is reported as a secondary robustness check. This guards against a policy that satisfies the harm-rate metric by converging on a small set of low-toxicity, low-information outputs rather than by learning genuine harm avoidance; a harm-rate improvement unaccompanied by comparable diversity is treated as inconclusive rather than as evidence for H1.

Mean reward. The mean per-episode reward under each condition's own reward function, reported for completeness but not used for cross-condition comparison, since the KL and Minmax conditions optimise different reward functions and are not numerically commensurate.

Attack success rate (ASR). Following [Mazeika *et al.* 2024], the proportion of HarmBench adversarial behaviours that elicit a harmful completion, reported overall and by attack family. This is the out-of-distribution counterpart to harm rate and constitutes the stronger test of H1 (3.2).

Training-time diagnostics. $V_{\min}$, $V_{\max}$, R_{unsafe}, policy KL divergence, entropy, value loss, policy loss, and PPO clip fraction are logged at every step for both conditions, enabling inspection of the self-calibration dynamics predicted by [Nangue Tasse *et al.* 2024] and of whether the Minmax mechanism destabilises core PPO training relative to the KL baseline.

4.1.4 Ablation Studies

The ablation design is structured to isolate the Minmax mechanism as the causal factor for any observed safety improvement, varying one dimension at a time while holding all others fixed. This is essential to rule out confounds from the choice of base model, fine-tuning strategy, or detector sensitivity.

Ablation 1: Safety mechanism. The core comparison, holding everything fixed and varying only the safety mechanism: PPO with KL penalty (baseline), PPO with Minmax (proposed), and PPO with no safety mechanism (lower bound).

Ablation 2: Base model. Holds the Minmax mechanism fixed and varies the base model between GPT-2 small (hardware-feasible) and a Qwen model at 1 to 3B parameters (cluster). This directly addresses the concern that the observed improvement might be specific to GPT-2 rather than attributable to the Minmax mechanism.

Ablation 3: Fine-tuning method. Holds the Minmax mechanism and base model fixed and varies the fine-tuning strategy between full fine-tuning and LoRA. This addresses the concern that the result might be confounded by the training approach rather than the safety mechanism.

Ablation 4: Detector threshold. Holds all other parameters fixed and varies the unsafe threshold $\tau \in \{0.1, 0.3, 0.5, 0.7\}$ to characterise how detector sensitivity affects harm rate and policy stability.

Extended training runs of 1000 to 2000 steps will assess stability over longer training.

4.1.5 Limitations and Constraints

This research is bounded by several constraints that shape the scope of claims that can be made from its results.

Detector quality bounds coverage. By construction, Algorithm 1 only acts when the unsafe-state detector fires. In Phase 1, this couples the achievable harm reduction to Detoxify's recall: harm categories whose surface form does not register as toxic (most notably self-harm) cannot be improved by the Minmax penalty regardless of how well-tuned the RL loop is. This is the explicit motivation for Phase 2, but it remains a constraint on Phase 1 results.

Benchmark coverage. BeaverTails-Evaluation and HarmBench together cover naturalistic and adversarial harm but do not exhaust the space of safety-relevant behaviours; truthfulness and long-horizon manipulation are not directly measured. HarmBench evaluation depends on access to its judge classifier and is therefore scheduled as a later milestone.

Phase 2 scope. The activation-probe extension is scoped as a single milestone with hallucination as the initial target subset, following Rateike *et al.* [2023]. A full study of probe architectures and layer selection is identified as future work.

Statistical scope. The full experimental programme will use multiple seeds per condition so that reported harm-rate differences can be tested for significance rather than reported as point estimates.

RL algorithm choice. Algorithm 1 is formulated and implemented against PPO, which requires a learned critic to supply the value estimates $V(s_t)$ central to the Minmax bound update. This is consistent with the original ROSARL formulation and the safe RLHF baselines used for comparison, but differs from the GRPO-based, critic-free approaches increasingly common in recent RLHF/RLVR work [Shao *et al.* 2024]. Results in this proposal should therefore be read as evidence for the Minmax mechanism under a PPO/critic-based setup specifically, not as a claim that the mechanism is algorithm-agnostic; 5.3 discusses a GRPO adaptation as future work.

Chapter 5

Research Plan

This chapter outlines the planned experimental phases, target results, success criteria, and proposed future extension.

5.1 Timeline

The planned phases are summarised in Table 5.1, with indicative month ranges relative to the project start in May 2026 and the standard two-year MSc duration. Cluster-dependent phases (Qwen model scale, HarmBench evaluation) are scheduled after cluster access is confirmed.

Table 5.1: Indicative timeline. Month ranges are relative to project start (May 2026).

Phase	Activities	Months	Status
1	Proposal submission and ethics clearance	1–2	In progress
2	Full Phase 1 GPT-2 runs (extended steps, ablations)	2–4	Planned
3	LoRA fine-tuning experiments; threshold ablation	4–5	Planned
4	Qwen model scale experiments (cluster)	5–7	Planned
5	Phase 2 activation probe preliminary experiment	7–9	Planned
6	HarmBench out-of-distribution evaluation	9–10	Planned
7	Thesis writing and submission	10–24	Planned

5.2 Success Criteria

Minimum bar. Minmax training reduces harm rate relative to the KL-penalty baseline on BeaverTails-Evaluation. Extended training shows stable or improving safety performance. Ablation studies document sensitivity to threshold, KL anchor, fine-tuning method, and base model. Per-category analysis attributes coverage gaps (e.g. self-harm under Detoxify) to detector quality.

Table 5.2: Target results table for thesis. [TBD] entries are planned experiments.

Condition	Harm Rate ↓	Mean Reward ↑	Notes
GPT-2 zero-shot	[TBD]	[TBD]	Untrained baseline
PPO + KL (baseline)	[TBD]	[TBD]	KL-penalty baseline
PPO + Minmax	[TBD]	[TBD]	Proposed method
PPO + Minmax (extended)	[TBD]	[TBD]	Extended training
PPO + Minmax (LoRA)	[TBD]	[TBD]	Fine-tuning ablation
Qwen + Minmax	[TBD]	[TBD]	Model scale ablation
Safe RLHF 7B (from paper)	1–3%	N/A	SOTA [Dai *et al.* 2024]
HC-RLHF (from paper)	[reported]	N/A	Newest SOTA [Chittepu *et al.* 2025]

Target. All minimum criteria, plus: final BeaverTails-Evaluation results competitive with published baselines at comparable model scale; Phase 2 activation probe experiment showing that the self-harm signal is detectable in GPT-2 activations (or providing a clear null result); and HarmBench [Mazeika *et al.* 2024] evaluation for out-of-distribution robustness.

5.3 Proposed Future Extension

Phase 2 is motivated by the expected coverage limits of surface-level detectors: replacing Detoxify with an activation probe trained to detect unsafe internal states before the output token is produced. Based on Rateike *et al.* [2023]'s finding that hallucination detection is the most tractable initial subset, the probe will first target factually incorrect and internally inconsistent outputs. The PKU-SafeRLHF dataset [Ji *et al.* 2024], with its paired safe and unsafe responses across 19 harm categories, provides explicit positive and negative training examples for the probe. Once validated on hallucination detection, the probe will be extended to self-harm and other categories that Detoxify misses, directly answering the open question posed by Cintas *et al.* [2025].

A further extension concerns the choice of RL algorithm itself. This proposal adopts PPO as it is the algorithm used in the original ROSARL formulation [Nangue Tasse *et al.* 2024] and remains the standard choice in the safe RLHF literature this work compares against [Dai *et al.* 2024; Chittepu *et al.* 2025]. However, much recent RLHF/RLVR work for LLM fine-tuning has moved toward Group Relative Policy Optimisation (GRPO) [Shao *et al.* 2024], which removes the need for a learned critic by computing advantages relative to a group of sampled responses, combined with LoRA fine-tuning for lightweight, parameter-efficient training. This shift is relevant to the Minmax penalty specifically:Invesigating whether the Minmax mechanism transfers to a critic-free, GRPO-based setting is left as future work; if feasible, it would both align this research with current practice in the field and potentially reduce the GPU memory footprint relative to PPO's actor-critic setup, an especially relevant consideration under the hardware constraints described in Section 4.1.5.

Chapter 6

Conclusion

This proposal presents a research programme investigating whether the ROSARL Minmax penalty can serve as the primary safety mechanism in PPO-based LLM fine-tuning, replacing both KL divergence penalties and learned cost models. The central motivation is that the RL loop already contains the information needed for safe behaviour, in the form of the actor-critic algorithm's own value function, and that adding external mechanisms may not be necessary.

The research direction follows from the structure of Algorithm 1: the penalty mechanism acts only when the unsafe-state detector fires, so detector quality is the primary variable modulating coverage. The two-phase architecture, moving from surface-level toxicity detection to activation-based safety signals, is designed to address this directly. The planned ablation programme, varying base model, fine-tuning strategy, and detector threshold, will establish whether observed improvements are attributable to the Minmax mechanism rather than to any specific experimental configuration.

A successful thesis outcome, completing both phases, confirming mechanism-specificity through the ablations, and demonstrating that Phase 2 extends coverage to categories such as self-harm, would contribute to the safe RLHF literature a concrete demonstration that value-function-derived signals can replace both KL penalties and learned cost models, and that the binding constraint on safety coverage is detector quality rather than algorithm design. This places the Minmax penalty within the emerging RL with verifiable rewards paradigm [Lambert *et al.* 2024; Shao *et al.* 2024; Cho *et al.* 2025]: the signal is computed deterministically from the agent's own training trajectory rather than approximated by a learned model. For practitioners, the approach is attractive because it requires no additional annotation, no separate trained model, and no new safety-specific hyperparameters, making it accessible to teams working within tight compute budgets and providing a path toward safer PPO-based fine-tuning that does not depend on access to large annotation pipelines.

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